When does Translation Help? Representation, Noise, and Adaptation in Low-Resource Multilingual QA

Franck Fongang

Motivation

Large language models (LLMs) are often described as multilingual, yet their performance in low-resource African languages remains uneven. Prior work has shown that data quality and representation are central bottlenecks, and that translated data can improve multilingual training.

The central question is:
<u>When does translation improve QA performance, and when does it introduce noise that degrades it in the presence of sufficient multilingual representation?</u>

Experimental Setup

We evaluate extractive question answering using the AfriQA dataset across multiple African languages (e.g., Hausa, Yoruba, Igbo, Zulu, Bemba, Fon).

Two pipelines are compared:
- Direct: Source-language question and context → LLM → answer in source language
- Translate-pivot: Source → English/French (NLLB) → LLM → answer → back to source

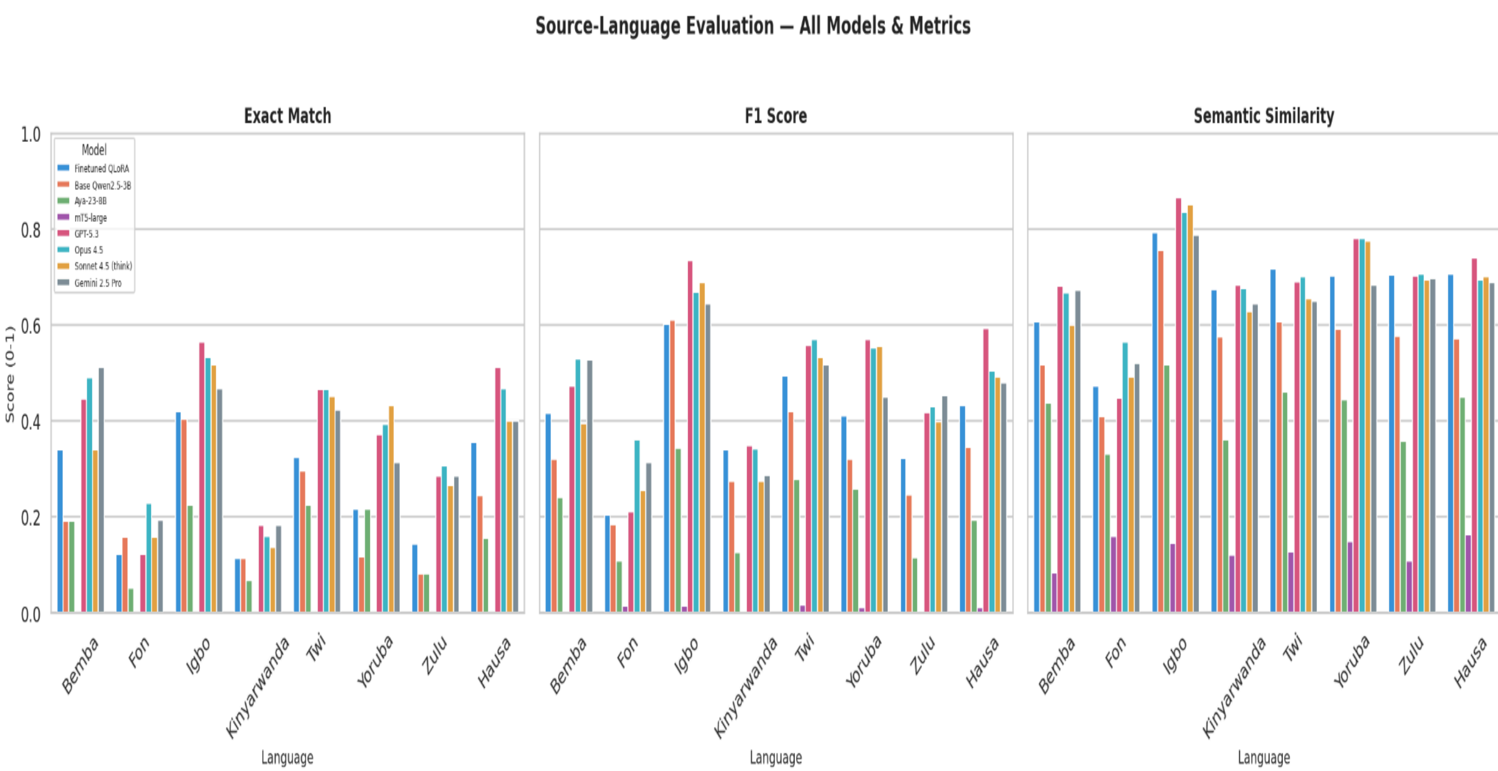
We benchmark:
- Frontier LLMs (GPT-5.3, Claude Opus/Sonnet (Thinking), Gemini)
- Multilingual baselines (Aya-23-8B, mT5)
- A locally fine-tuned Qwen2.5-3B (QLoRA) model

Metrics:
- Exact Match (EM)
- Token-level F1
- Semantic similarity (LaBSE)

Key Findings

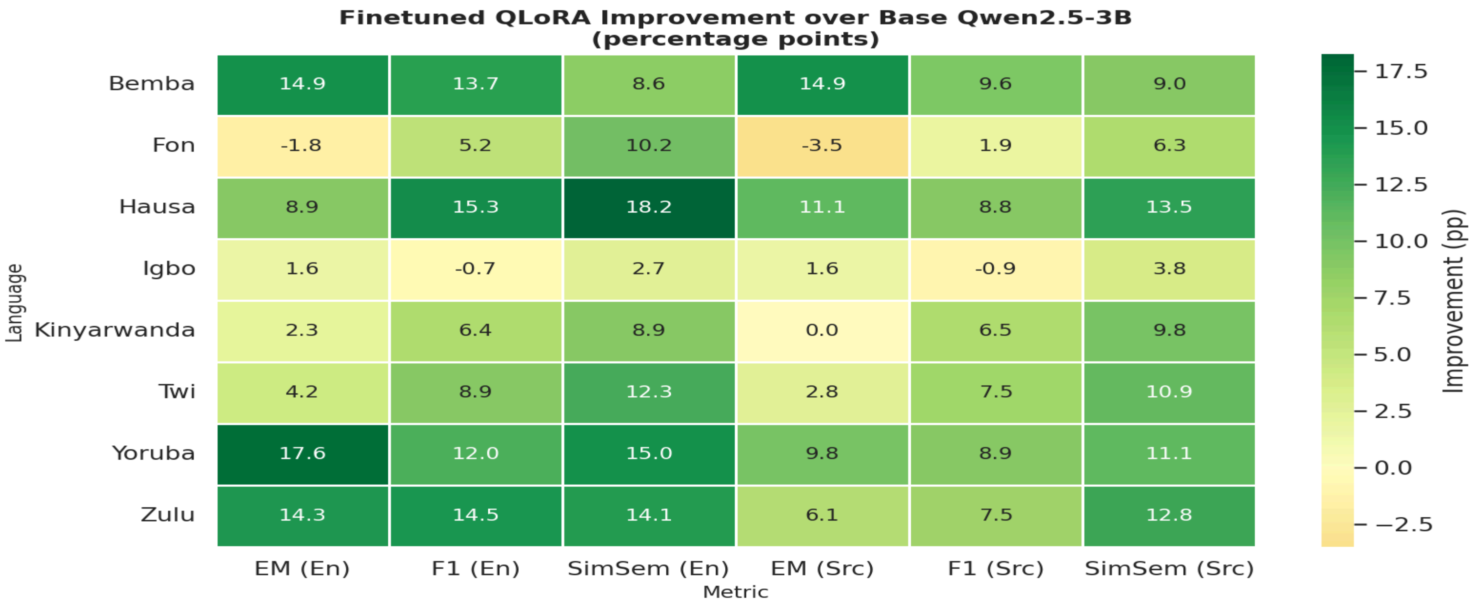
- Translation degrades performance for strong models
  For frontier LLMs, translate-pivot introduces consistent drops (~-0.19 to ~0.24 EM), driven by compounding translation errors.
- Direct generation is viable in higher-resource languages
  Models achieve strong performance directly in languages such as Igbo and Hausa, suggesting non-trivial multilingual representation.
- Translation remains useful for weaker or unadapted models
  Multilingual baselines benefit more from translation, indicating insufficient internal representation.
- Training on translation-induced noise improves robustness
  A QLoRA-adapted 3B model trained on machine-translated and round-trip corrupted data improves performance across several languages, particularly lower-resource ones, outperforming larger multilingual baselines in multiple cases.

Figure 1 - Source-language performance (Direct vs Translate-Pivot)



Translation consistently degrades performance for frontier models while benefiting weaker ones.

Figure 2 - Effect of noise-aware fine-tuning



Training on machine-translated and round-trip data improves robustness and closes performance gaps.

Interpretation

These results suggest that translation plays a conditional role in multilingual QA :

- As an alignment mechanism, it helps when models lack sufficient representation
- As a noise source, it degrades performance when representation is already strong

We hypothesize that:
<u>Translation improves QA performance only when representation is insufficient, and otherwise introduces noise that degrades performance.</u>

Interestingly, the fine-tuned model is trained on imperfect, machine-translated data rather than clean supervision. This suggests that training on translation-induced noise improves robustness to multilingual pipeline artifacts, allowing the model to better handle imperfect inputs.

Discussion

These findings raise several questions aligned with recent work on multilingual evaluation and adaptation :

- When does translation help vs hurt in multilingual inference?
- How does representation quality vary across languages within a single model?
- Can targeted adaptation eliminate the need for translation pipelines?

Understanding this trade-off is critical for building systems that operate natively in low-resource languages without relying on translation as an intermediary. These results suggest that improving representation through targeted adaptation may be more effective than relying on translation-based pipelines, especially as model capacity increases. This suggests a shift from translation-based pipelines toward representation-centric approaches for low-resource multilingual QA.

Conclusion

Translation is not universally beneficial in multilingual QA. While it can compensate for weak representation, it introduces noise that degrades performance for stronger models. Improving representation quality through better data and targeted adaptation may therefore be a more effective direction than relying on translation-based pipelines.

Code

https://github.com/Francklin9999/low-resource-qa-representation